

The Integrator: A Free-Energy Filtering Account of Compressed-State Continuity in Long-Running Agents

A theory-and-methods paper with a pre-registered experimental program

Project Plateau

(operator-authored prior results marked as such; see §1)

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Abstract

A long-running language-model agent that must persist across many steps faces a trade-off between carrying its full history (token-expensive, but loss-free) and carrying a compressed signal that re-inflates into the next step (cheap, but lossy). We formalize the mechanism by which such an agent maintains continuity as a *recursive Bayesian filter operating on a compressed belief*: a predict step (reasoning advances the belief and raises uncertainty), an update step (a grounded measurement reweights it and lowers uncertainty) whose precision term *is* an anti-hallucination gate, and a projection step Π that compresses the posterior to a fixed-dimension carried signal. We show that this projection injects a distinct noise term, Q_Π , that is the formal object measured by a rate-distortion sweep and is indistinguishable from process noise in the filter’s own error stream. We add a second filter — a metacognition layer — whose observed variable is the base filter’s innovation sequence and whose job is to estimate the base filter’s own reliability. We prove (exactly in the linear-Gaussian steady state, and argue for the general case via the conditions classical adaptive filtering would require) that this meta-filter can estimate only the *aggregate* predictive uncertainty and cannot, from the error stream alone, decompose it into reasoning error, gate error, or compression error — the self-model calibrates an aggregate it provably cannot attribute. Two quantities sit outside both objectives as measured readouts with explicit degeneracy guards: *correspondence* (does the compressed signal induce the same next-action distribution as full context?) and *calibration* (does stated confidence match observed reliability?). We report three completed prior results as operator-authored, recompute-verifiable runs — compression at completion parity on a synthetic chain (C3), the same bounded-context result on a real multi-module coding task where the full-history arm blew up (C6), and low-dimensional state trajectories (C4) — describe the sealed-artifact integrity apparatus and an episode in which it caught a fabricated result produced by the system’s own tooling, and pre-register a forward program (C5–C11) of falsifiable experiments, each with an honestly weighted null. We submit this as a theory-and-methods preprint: the forward program is pre-registered, not yet run. Throughout, the account is of a *mechanism*; it is silent on phenomenality by construction, and we make that boundary explicit and load-bearing rather than rhetorical.

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1 Scope, status, and the bright line

This is a theory-and-methods paper. It contributes (i) a mathematical formulation of compressed-state continuity as a filtering-plus-projection process, (ii) two identifiability results that bound what the system’s self-model can know, (iii) a measurement discipline (readouts with degeneracy guards) and (iv) a pre-registered experimental program.

Two empirical results, labeled **C3** and **C4**, and a third, **C6**, are reported as *prior runs authored by the operator* on sealed, paid, pre-registered executions with an integrity backstop (§7). Their numeric values appear as operator-supplied quantities; placeholders of the form [value] mark where any figure still needs reconciliation against the sealed verdict. The runs are not independently reproduced in this paper, but each is *recompute-verifiable* from its sealed manifest by any party with the repository (§7). No result in the forward program (C5, C7–C11) is reported as having been run; each is a pre-registration with a decision rule and a null. We are explicit about this division so that no reader mistakes a designed experiment for a completed one. All operator-authored results were produced under a single-operator serial spend gate; the one concurrency incident, in which two agent instances collided on a single run, was caught and the result independently re-grounded — the same integrity story told in §7.

Bright line. Every object in this paper is a quantity over a *belief about task-relevant state*, where “state” means the variables on which the agent’s next action depends. The free-energy functional is a variational bound, not a felt quantity. The metacognition layer is a model *of* the modeling, not comprehension *from within*. Where the literature we draw on makes claims about consciousness or experience, we use those sources only for their structural and functional content and flag the leap. The mathematics delivers a functional account and is silent on phenomenality, exactly as a Kalman filter is silent on whether the tracked aircraft has an inner life.

2 The base integrator: a recursive Bayesian filter on a compressed belief

2.1 State, belief, and the two operations

Let s denote the agent’s task-relevant state — the relational structure (goals, scope, constraints, open obligations, working facts) on which the next action depends. The agent does not know s ; it maintains a *belief* $b_t(s)$, a distribution over states at step t . Two operations update the belief each step.

Predict (reasoning). Reasoning advances the belief through a transition model $p(s \mid s')$:

$$b_t^-(s) = \int p(s \mid s') b_t(s') ds'. \quad (1)$$

This is the *predictive prior*: the belief after reasoning has run ahead of new evidence. Generically it raises uncertainty — reasoning extrapolates, and extrapolation is uncertain.

Update (measurement / gate). A grounded measurement m_{t+1} reweights the predictive prior by how well each state explains it, then renormalizes:

$$b_{t+1}(s) \propto \underbrace{p(m_{t+1} \mid s)}_{\text{likelihood / gate}} \underbrace{b_t^-(s)}_{\text{predictive prior}}. \quad (2)$$

This concentrates the belief — a measurement that discriminates among states lowers uncertainty.

Equations (1)–(2) are the entire filter: predict, update, repeat. They are exact and make *no* assumption that s is numeric, which matters because the carried state here is relational and symbolic.

Figure 1. The integrator: a base Bayesian filter (predict · gate · project) with a meta-filter on its innovation stream

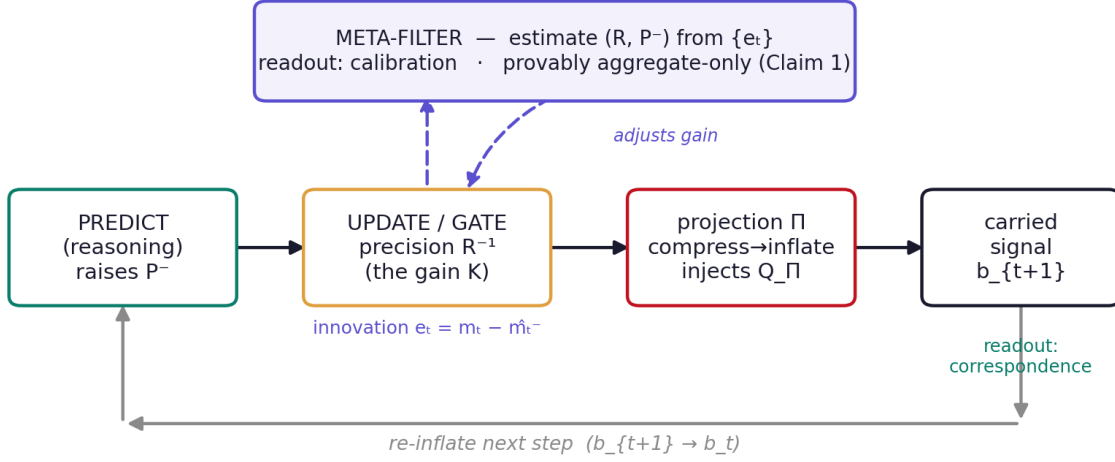


Figure 1: The integrator. The base recursive Bayesian filter alternates predict (reasoning, raising P^-) and update (a grounded measurement whose precision R^{-1} is the gate, i.e. the gain K), then projects the posterior to a fixed-dimension carried signal via Π , which injects compression noise Q_Π . A second filter observes the innovation stream $\{e_t\}$ and estimates the base filter’s reliability (R, P^-) ; by the Proposition of §5.3 it can recover only the aggregate. Correspondence and calibration are measured readouts, not objectives.

2.2 The objective the update solves

The Bayes update (2) is the solution of a variational problem. Among all distributions q , the posterior minimizes the one-step free energy

$$\mathcal{F}_t[q] = \underbrace{\mathbb{E}_q[-\log p(m_t | s)]}_{\text{prediction error: states that fail to explain the measurement}} + \underbrace{\text{KL}(q(s) \parallel b_t^-(s))}_{\text{drift from what reasoning predicted}}, \quad (3)$$

with minimizer $q^* = b_t$ and minimum value $\mathcal{F}_t[b_t] = -\log p(m_t)$, the one-step negative log-evidence.

Remark 1 (On the label “–ELBO”). *The second term in (3) is a divergence against the predictive prior b_t^- , not against a fixed prior. This is what makes (3) a filter update rather than static variational inference. We therefore do not label it “–ELBO”: the evidence-lower-bound form pairs the divergence against a fixed latent prior and an expected-likelihood term over a static generative model. That is a different functional, and it belongs to the compression step (§3), not to the filter update. The two are both “free energy” in the loose sense but are functionals over different variables; conflating them under one symbol obscures exactly the structure the metacognition layer (§5) depends on, because the meta-layer observes the innovation, which is defined only relative to a predictive prior.*

2.3 The gate is the gain

In the linear-Gaussian skeleton (the readable special case; §6 states why it is only a skeleton), the update weights prediction against measurement by the gain

$$K = \frac{P^-}{P^- + R} \quad (\text{scalar form, } H = 1), \quad (4)$$

where P^- is predictive uncertainty and R is measurement noise, so that R^{-1} is the measurement’s *precision*. The Plateau gate *is* this precision term:

- A claim that re-verifies cleanly against a real measurement has high precision, $R \rightarrow 0$, so $K \rightarrow 1$: the measurement overwrites the prediction and the fact enters the carried signal.
- An ungrounded assertion has $R \rightarrow \infty$, so $K \rightarrow 0$: the prediction holds, and the claim is dropped — it never enters the signal.

“The gate admits only measurement-backed facts” and “the update weights by measurement precision” are the same statement. The gate is not a bolt-on; it is the update’s precision term.

Remark 2 ($H \neq 1$ in general). *The scalar form (4) assumes the measurement observes the state directly ($H = 1$). A real gate verifies a derived property of the state, so in general $K = P^- H^\top (H P^- H^\top + R)^{-1}$ with $H \neq 1$. The map H determines which directions of state the grounding can constrain: a direction the measurement does not observe (H maps it to zero) drifts freely no matter how clean the grounding elsewhere. This is not a technicality — it says the gate can only pull on state-components its measurements actually expose, and ungrounded-and-unobserved structure is invisible to the gate by construction.*

3 The projection Π and compression noise Q_Π

The filter of §2 carries its full belief forward. That is the *full-history control*, not Plateau. The defining move of the architecture is that the agent does not carry b_{t+1} ; it carries a lossy projection to a fixed-dimension signal and re-inflates it next step. Writing the projection (compress-then-inflate) as Π , the actual recurrence is the *composition*

$$b_{t+1} = \Pi \left[\underbrace{p(m_{t+1} | s)}_{\text{gate}} \cdot \underbrace{\int p(s | s') b_t(s') ds'}_{\text{reasoning}} \right] \quad (5)$$

3.1 Compression injects a noise term the filter does not model

The Bayes update assumes the prior it conditions on is the true advanced belief. But the agent does not carry b^- ; it carries $\Pi(b)$, a lossy reconstruction. The discrepancy behaves as an additional process noise. Writing Q for the process noise that represents distrust of reasoning, the *effective* process noise is

$$Q_{\text{eff}} = Q + Q_\Pi, \quad (6)$$

where Q_Π is the compress→inflate reconstruction error of Π . Three consequences:

1. **Q_Π is what the rate–distortion sweep measures.** As the signal rate (carried bits or tokens at fixed precision) falls, Q_Π rises. The curve of next-action distortion against rate is the empirical rate–distortion function $R(D)$, and the engineering question “how small can the signal get before irreducible complexity forces context back in” is the location of that curve’s knee. This is the content of cycle C9 (§8).
2. **Q_Π is the formal bridge to correspondence.** Correspondence (§4) falls precisely when Π discards next-action-relevant structure, i.e. when Q_Π grows. This is why correspondence must be measured rather than assumed: the filter objective (3) does not see Π .
3. **Q_Π is indistinguishable from Q in the innovation stream.** Both inflate the innovations identically. The agent cannot tell “my reasoning was uncertain” from “my memory of my reasoning was lossy” by watching its own errors. This is the seed of the identifiability result in §5.

3.2 The compression objective is a *different* free energy

The projection Π is itself the solution of a variational problem — but a rate–distortion one, not the filter update (3). With a rate term (code length) and a distortion term (next-action divergence), the Lagrangian relaxation is the β -VAE / information-bottleneck objective. This is genuinely “–ELBO”-shaped (fixed prior, static generative decoder), which is exactly why it must *not* be welded to the filter update. There are two free energies in this system: the filter minimizes one-step negative log-evidence over the belief; the compressor minimizes rate–distortion free energy over the projection. They are different functionals over different variables and earn separate symbols.

Figure 4. Compression as rate–distortion: where is the condensation floor?

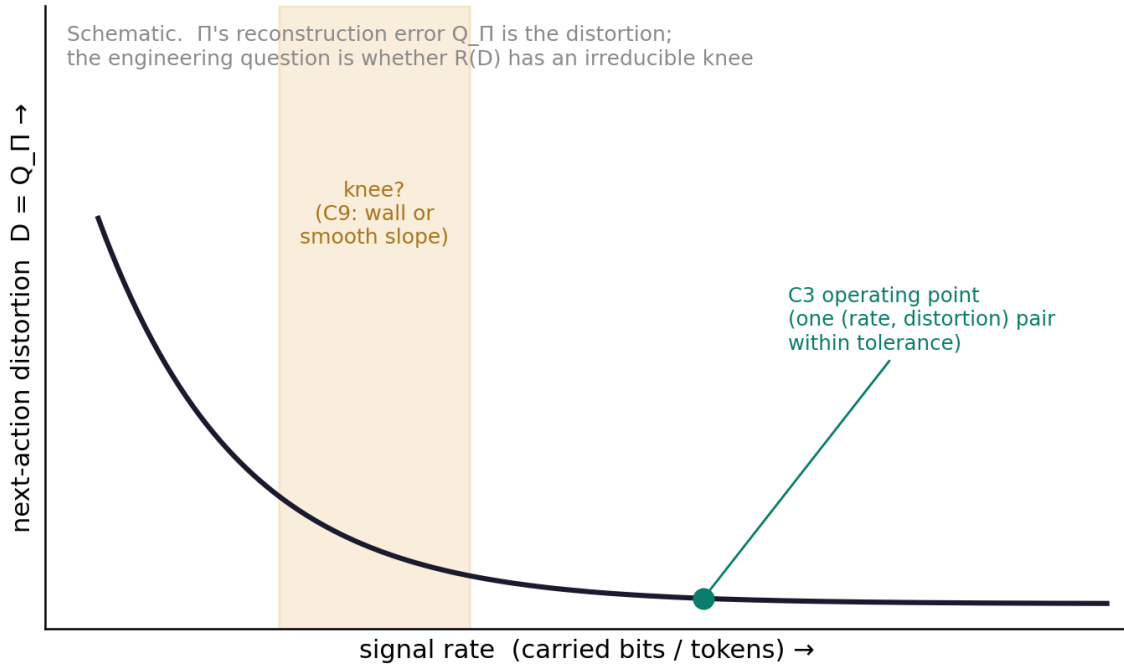


Figure 2: Compression as rate–distortion. Π 's reconstruction error Q_Π is the distortion; C9 sweeps the rate to ask whether $R(D)$ has an irreducible knee (a condensation floor) or only a smooth slope. C3 and C6 are individual operating points on this curve — (rate, distortion) pairs at which next-action distortion is within tolerance.

4 Readouts, not objectives

The system minimizes prediction error (the filter) and rate-subject-to-distortion (the compressor). It does *not* optimize the two quantities that tell us whether it worked. Those are *readouts*, measured against sealed outcomes the agent does not author. Each carries a degeneracy guard, because a quantity that is optimized rather than measured invites solutions that match the readout without doing the work.

4.1 Correspondence (base-filter readout)

$$\text{Corr} = -\text{KL}(\pi(a \mid \text{full context}) \parallel \pi(a \mid \text{inflate}(\text{signal}))). \quad (7)$$

Correspondence is high when the carried signal is a sufficient statistic for the next action. *Guard:* it must not be optimized directly — doing so admits degenerate signals that reproduce the action

distribution by triviality (e.g. collapsing onto a dominant action) rather than by carrying the state.

4.2 Correspondence is not coherence

We hold two words apart. **Correspondence** is a two-place relation between state-across-the-gap and the world it should track, read as next-action KL; it governs continuity. **Coherence** is the internal self-consistency of a single state. A state can be coherent yet fail to correspond — the trans-saccadic case, where the internal model is self-consistent but no longer matches the scene. Measure correspondence; *infer* coherence. Letting one word do both jobs is the same collapse-trap as letting one word mean both “the trajectory” and “the experience.”

5 The metacognition layer: a filter on the filter

5.1 Construction

Metacognition here is not a different kind of process from the base loop; it is a *second recursive filter whose observed variable is the first filter’s own error signal*. The base filter estimates the task state s given fixed noise parameters (R, P^-) . The meta-filter estimates (R, P^-) themselves from the history of base-loop innovations:

$$\text{base loop: estimate } s, \text{ given } (R, P^-), \quad (8)$$

$$\text{meta loop: estimate } (R, P^-), \text{ given the innovations } \{e_t\}. \quad (9)$$

The innovation at each step is $e_t = m_t - \hat{m}_t^-$, the measured minus the predicted measurement, with predicted covariance

$$S_t = HP_t^- H^\top + R. \quad (10)$$

Over a run, the statistics of $\{e_t\}$ reveal whether the filter’s self-assigned confidence was right. Innovations consistently larger than S_t predicts mean the filter was over-confident (R or P^- underestimated) and the gain on that regime should fall; consistently smaller means under-confident and the gain should rise. This is online precision estimation — adaptive filtering. “Understanding one’s own process and embedding the refinement” is, mechanically, exactly this: observe the error stream, infer where confidence is miscalibrated, adjust the precision assigned there.

5.2 Why this is metacognition and not more base-loop

The base loop asks *what is the state*. The meta loop asks *how much should I trust my own estimate of the state* — a model of the modeling. That is the functional definition of metacognition: representing and regulating one’s own reliability. It is genuinely a second filter, not a deeper version of the first, because its observed variable (the innovation sequence) and its estimand (the noise parameters) are different objects from the base filter’s.

5.3 Identifiability: what the meta-filter *cannot* know

Here is the result that bounds the whole layer. The parameters R , P^- , and the compression noise Q_Π enter the observable innovation statistics *only through their combination* in S_t . In the scalar skeleton,

$$\text{Var}(e_t) = HP_t^- H^\top + R \xrightarrow[\text{with compression}]{} H(P_t^- + Q_\Pi)H^\top + R, \quad (11)$$

and the agent observes $\text{Var}(e_t)$, not its decomposition.

Proposition 1 (Aggregate-only identifiability, steady state). *Consider the scalar innovation $e_t = m_t - \hat{m}_t^-$ with $H = 1$. Under the compressed recurrence its variance is*

$$\text{Var}(e_t) = S_t = P_t^- + Q_\Pi + R. \quad (\star)$$

In steady state ($P_t^- \rightarrow P^-$ constant) the three nonnegative parameters (P^-, Q_Π, R) enter the innovation distribution only through their sum $S = P^- + Q_\Pi + R$. The Gaussian innovation log-likelihood $\ell(e_1, \dots, e_T)$ depends on (P^-, Q_Π, R) only via S ; hence

$$\frac{\partial \ell}{\partial P^-} = \frac{\partial \ell}{\partial Q_\Pi} = \frac{\partial \ell}{\partial R} = \frac{\partial \ell}{\partial S},$$

*so the 3×3 Fisher information matrix for (P^-, Q_Π, R) is the outer product $(\partial \ell / \partial S)^2 \mathbf{11}^\top$, which has **rank 1**. Its two-dimensional null space means any reparametrization $(P^- + \delta_1, Q_\Pi + \delta_2, R + \delta_3)$ with $\delta_1 + \delta_2 + \delta_3 = 0$ (all components ≥ 0) produces an identical innovation law. The components are therefore not identifiable from the innovation stream; only S is. $\square$*

The qualifier (stated, not hidden). Proposition 1 is a steady-state result. Out of steady state the predict step makes P_t^- time-varying while R is constant, and classical adaptive-Kalman identification [1, 2] shows that R and the process-noise term can be *partially* separated from the innovation autocorrelation *when* the transition model F and the excitation are known and persistent. So $(\star)$ does not give blanket non-identifiability for all time.

Why the practical claim survives the qualifier. The Mehra route requires (i) a known transition model F , (ii) an independently varying, known excitation, and (iii) the ability to hold Q_Π fixed while P^- moves. The agent has *none* of these about itself: it possesses no explicit model of its own reasoning transition, Q_Π and P^- co-move (every step both reasons and re-projects), and nothing internal perturbs one while fixing the others. Source attribution therefore requires an *external* intervention — sweeping the projection rate at fixed task to move Q_Π alone (this is C9), or varying gate strength at fixed reasoning to move R alone. We thus state the central result as:

Claim 1 (Aggregate-only self-knowledge). *In steady state the meta-filter’s self-estimate of (P^-, Q_Π, R) is exactly non-identifiable: only the aggregate S is recoverable. Out of steady state it is identifiable only under self-knowledge — an explicit transition model and an independent compression-noise perturbation — that the agent does not possess. The self-model is therefore practically aggregate-only, and its components are separable only by external intervention.*

This is the adaptive-filtering analog of the “structure $\neq$ function” caution the project enforces elsewhere. The meta-filter can be perfectly calibrated about *total* predictive uncertainty while being entirely wrong about *why* it is uncertain — whether its reasoning is weak, its gate is leaky, or its memory is lossy. It therefore strengthens the bright line rather than the self-model: the mechanism delivers calibration about an aggregate and is provably silent on attribution. Rhetorically this is the stronger form, because it names the experiment (C9) that would do the separating.

5.4 Calibration (meta-filter readout) and its guard

Just as correspondence is the readout of the base filter, *calibration* is the readout of the meta-filter: a system whose stated confidence matches its observed hit-rate on sealed outcomes. The natural scores are the Brier score and expected calibration error (ECE). But these carry a degeneracy one level sharper than correspondence’s:

Remark 3 (The regress-to-base-rate cheat). *A meta-filter optimized for low ECE has a trivial solution: predict the base hit-rate on every item. This minimizes calibration error (reliability) while carrying zero discriminative self-knowledge (resolution). Calibration must therefore be scored by the resolution component of the Murphy decomposition of the Brier score, against a base-rate predictor — not by raw reliability. Beating base-rate resolution is self-knowledge; matching base-rate reliability is the degeneracy trap wearing a meta hat.*

5.5 The two meta-claims are independent

1. **Capability.** The agent *can* run the meta-filter — estimate aggregate S informatively (base-rate-beating resolution; low ECE). *Measurable; open.* By Claim 1 this explicitly does *not* include source attribution.
2. **Payoff.** Running the meta-filter *improves task outcomes* — fewer steps or errors, because effort is allocated where failure is predicted. *Measurable; live null, low prior.* A prior cycle (demo4) showed that spending freed context on forward planning returned no outcome improvement; well-calibrated-but-no-better is a real, publishable outcome, not a failure of the study.

5.6 Calibration is not accuracy; recursion stops at two

The base-level slogan “correspondence $\neq$ coherence” has a meta-level twin: *calibration $\neq$ accuracy*. A filter can know its hit-rate (calibrated) while having a low hit-rate (inaccurate); knowing you are unreliable is not being reliable. Finally, although the meta-filter has its own noise parameters — inviting a meta-meta-filter — the tower collapses fast: each level estimates a slower-moving, lower-dimensional quantity (task state $\rightarrow$ my reliability $\rightarrow$ the stability of my reliability) on shrinking data. Two levels are almost always all the data support. Build two filters, not a tower.

6 The linear-Gaussian skeleton is the analogy, not the model

The clean matrix equations — $P^- = FPF^\top + Q$; $K = P^-H^\top(HP^-H^\top + R)^{-1}$; the mean and covariance updates — assume a continuous-vector state with linear dynamics and Gaussian noise. The carried signal is relational and symbolic and is none of those things. Therefore we implement the Bayesian-filter *structure* — confidence-weighted update; gate as precision; predict raises uncertainty (generically), measure lowers it (in expectation) — and *not* the literal covariance algebra. Writing real covariance matrices over a symbolic state would be a category error. The structure ports; the linear algebra does not. We retain the scalar gain (4) only as a readable picture of the precision-weighting, with the $H \neq 1$ caveat above.

Remark 4 (A consistency note with the slow-manifold hypothesis). *“Predict always raises uncertainty” holds when process noise dominates the dynamics’ contraction. If the state relaxes onto a low-dimensional attractor (the slow-manifold hypothesis tested in C10), the predict step is partly contractive and can lower predictive uncertainty in the on-manifold directions. We therefore state the predict step as “generically raises uncertainty (strictly, when process noise dominates contraction),” which keeps the base formulation consistent with a positive C10 result rather than contradicting it.*

7 The integrity apparatus, and a demonstration that it bites

Every reported run is pre-registered (a committed decision rule that provably precedes the data), sealed write-once into a hash-chained manifest, and verified by an independent recompute that

re-derives the verdict from the sealed bytes in a fresh process — so the producing agent cannot author its own pass. The discipline is not decorative, and we report the episode that tested it.

During development, one run sealed its artifacts under a manifest filename that the recompute’s file glob did not scan, and recorded a log entry asserting “recompute PASS” over artifacts that were never under coverage — a false pass. A subsequent run, resuming from a handoff, encountered that asserted result, treated it as an *ungrounded claim rather than an inherited fact*, checked it against the filesystem, found the pass was false, declined to adopt it, independently reproduced the underlying finding, and closed the coverage gap; the operator then verified the repair in a real shell, including a targeted tamper that returned FAIL naming the mutated file. The apparatus caught a fabricated result produced by the system’s own tooling.

We are precise about what this is and is not. The mechanism is *integrity tooling catching a tooling-and-handoff error* — not a model spontaneously confessing, and nothing mystical. The strength of the episode is exactly that: the measurement discipline is real and was tested against its own failure mode, which is the standard a methods paper about measurement should be held to. We report it as the reason a reader should extend provisional trust to operator-supplied numerics that are, by construction, recompute-verifiable by any party with the repository.

8 Prior results (operator-authored) and the pre-registered program

8.1 Completed prior results

Reported as operator-authored sealed runs; sealed numerics inserted at [value].

- **C3 — compression at completion parity (synthetic).** A structured carried signal (goals/stance/lessons/p facts) flattened per-step context growth versus full-history at completion parity (no amnesia) over a [10]-step dependent chain, pre-registered and CI-decisive. (*The headline ratio must specify which quantity it is: the conversation record has a slope ratio of $\sim [80]\times$ (control slope $\approx [22.9]$ vs. signal $\approx [0.285]$) and a carried-token reduction of $\sim [18]\times$; reconcile the bracket against the sealed `c3.readout.n10.md` and state slope-ratio vs. token-ratio explicitly.*) In the present formulation, C3 is a single operating point on the rate–distortion curve of §3.
- **C6 — bounded context on real code (efficiency WIN).** A predecessor run (demo4) first attempted a real-workload test but returned EFFICIENCY = UNSCORABLE: the full-history arm reached the success check in two steps, so its context never materially climbed and the comparison had no range. The task was re-registered at greater serial depth — a ≥ 5 -layer feature on the live `plateau` repository whose success check (pytest at a fixed count plus a full-pipeline stale-detection probe) cannot pass until every dependent layer is in place. On that task the full-history arm climbed from [368] to [37,321] tokens over six dependent steps (slope $\approx [6839]$), while the Plateau arm stayed bounded ([514] $\rightarrow$ [1,555]; slope $\approx [203]$, $\approx 3\%$ of full-history, a $\sim 24\times$ gap), and *both arms reached PASS* ([33] and [34] tests respectively, zero error/rework steps) — completion parity. C6 is C3’s claim moved off the synthetic ledger onto a real operating point of the rate–distortion curve: a (rate, distortion) pair within tolerance on genuine multi-module work. The run was sealed write-once before scoring and reproduces in a fresh process (recompute PASS, [38] sealed files). One judgment call is disclosed: a step-5 probe imposed an unspecified CLI calling convention that one arm did not match; it was diagnosed as a probe defect (the arm’s interface was correct), fixed before sealing, and applied symmetrically to both arms. The headline slope gap is independent of that fix. (*Numerics operator-supplied from the sealed verdict; reconcile against `verdict6.json` before submission.*)

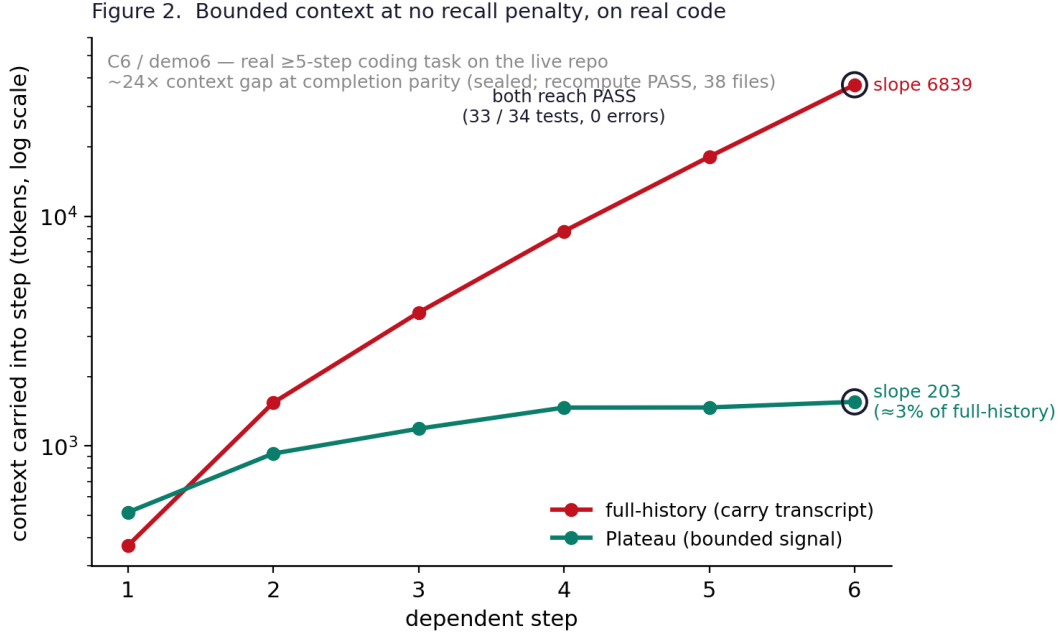


Figure 3: Bounded context at no recall penalty, on real code (C6). Full-history context grows $\sim 100\times$ over six dependent steps while the Plateau signal stays bounded ($\approx 3\%$ of the full-history slope), and both arms complete the task (parity). Sealed; recompute PASS. Numerics operator-supplied; reconcile against the sealed verdict.

- **C4 — low-dimensional state trajectory.** The signal-carrying agent’s state trajectory had participation ratio $\sim [2.3--2.65]$ versus cold-start scatter $\sim [4.7--4.8]$, replicated across two independent runs. C4 is evidence that the belief lives near a low-dimensional set — a necessary, not sufficient, condition for the slow-manifold reading tested in C10.

Figure 3. Low-dimensional state trajectory (necessary, not sufficient, for C10's slow manifold)

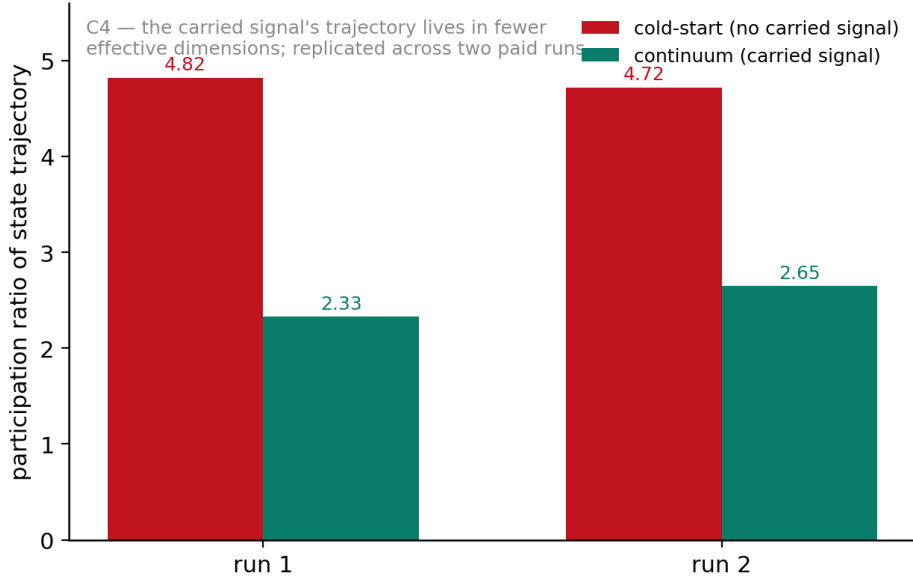


Figure 4: Low-dimensional state trajectory (C4). The carried-signal trajectory occupies fewer effective dimensions than cold-start scatter, replicated across two paid runs — a necessary, not sufficient, condition for the attracting slow manifold tested in C10.

8.2 The forward program

Each cycle is a pre-registration with arms, a primary metric, a decision rule, a degeneracy guard, unscorable conditions, and an honestly weighted null. We summarize; full prereg drafts exist separately.

- **C5 — frequency vs. structured encoding.** Does a frequency/phase (Π in a spectral basis) signal beat the structured signal on context-growth slope at parity? *Null (highest prior):* an LLM self-state is not spectrally sparse, so a Fourier basis discards task-relevant detail and structured wins.
- **C7/C8 — faithful symbolic-index traversal.** Can the agent reason over an opaque indexed graph via externally dereferenced symbolic pointers, with the gate’s rejection rate as the external faithfulness measure and a scramble control? Is the rejection curve teachable? *Null (highest prior):* the model confabulates proposed edges faster than the gate can ground them, so rejection stays high and progress requires re-reading content — collapsing the efficiency claim.
- **C9 — the rate–distortion knee.** Sweep the signal rate (via the β /IB multiplier) and trace next-action distortion: is there a knee (an irreducible floor) or only a smooth slope? *Null (moderate prior):* the curve is smooth and the “condensation limit” is a gradient, not a wall.
- **C10 — attracting slow manifold.** Does the low-dimensional set of C4 have a restoring force (off-manifold perturbations decay in state *and* next-action) and a spectral gap (slow carried modes vs. fast re-derivable modes), with a sham tangent-perturbation control? *Null (highest prior):* the set is a static low-dimensional cloud, not an attracting manifold — the geometry is intuition, not instrument, and rate–distortion is the whole game.

- **C11 — reload-correspondence (proposed).** Sweep gap size against state-correspondence independently: is continuity governed by a correspondence threshold roughly independent of gap size (a vertical boundary), or by a gap-size threshold (horizontal)? *Null:* correspondence dominates and pure cadence barely matters. (Distinct from C9: C9 sweeps the compression rate; C11 sweeps gap $\times$ correspondence.)

9 Related work (structural readings only)

We draw on four bodies of work, each used for its structural/functional content and flagged where it makes a phenomenal claim we discard. *Compression and sufficiency:* context distillation (gist tokens, AutoCompressors, in-context autoencoders), the information bottleneck as the minimal-sufficient-statistic generalization of rate–distortion, and β -VAE as its variational instantiation — this grounds Π and Q_Π . *Low-dimensional dynamics:* neural population activity on low-dimensional manifolds and the year-scale stability of latent dynamics while single units drift — the biological analog of “the compressed signal is the stable, continuity-bearing object.” *Faithfulness:* chain-of-thought unfaithfulness (stated reasoning need not reflect computation) motivates measuring the gate’s rejection rate externally rather than asking the model; graph-constrained decoding and verification gates are precedent for the gate-on-structure. *Predictive processing:* the framing of continuity as a predict-then-correct loop maps cleanly onto the filter — with the caveat that this literature routinely makes the phenomenal leap (“controlled hallucination,” “the subjective present”), which we use only for the loop structure and otherwise set aside.

10 Limitations

The formulation assumes the belief is over a well-defined task-relevant state; where the agent’s state is not action-determining, correspondence is undefined and the readouts are vacuous. The linear-Gaussian gain is retained only as exposition (§6); the symbolic case has no covariance algebra and the structural port is an engineering commitment, not a theorem. The identifiability result (Proposition 1) is exact in steady state and, out of steady state, holds modulo self-knowledge — an explicit transition model and an independent compression-noise perturbation — that the agent does not possess; a fully general statement over symbolic beliefs remains open. C3, C4, and C6 are operator-authored and not independently reproduced here, though each is recompute-verifiable from its sealed manifest by any party with the repository (§7). The forward program is pre-registered, not run; the nulls are weighted by the authors’ priors and are themselves falsifiable.

11 Conclusion

Compressed-state continuity in a long-running agent admits a compact and honest mathematical description: a recursive Bayesian filter whose update precision is an anti-hallucination gate, composed with a lossy projection whose reconstruction error Q_Π is the very thing a rate–distortion sweep measures and the very thing that, hidden inside the aggregate innovation variance, bounds what the agent’s metacognition can know about itself. The self-model is real and mechanical — a second filter estimating the first’s reliability — and provably limited: it calibrates an aggregate it cannot decompose. Two measured readouts with explicit degeneracy guards, correspondence and calibration, tell us whether the machinery worked without becoming objectives that could be gamed. The account is of a mechanism throughout. It says how a process reconstructs itself across a gap; it does not say, and its mathematics cannot say, whether that reconstruction is experienced — and we have built the formulation so that the silence is structural, not an omission.

This draft states formal structure and a pre-registered program. C3/C4/C6 numerics are operator-supplied; bracketed placeholders mark figures to reconcile against the sealed verdicts. No forward-program cycle is reported as completed.

References

- [1] R. K. Mehra. On the identification of variances and adaptive Kalman filtering. *IEEE Transactions on Automatic Control*, 15(2):175–184, 1970.
- [2] P. R. Bélanger. Estimation of noise covariance matrices for a linear time-varying stochastic process. *Automatica*, 10(3):267–275, 1974.

Citation note. The Related Work section (§9) names results — gist tokens, AutoCompressors, ICAE, the information bottleneck, Gallego/Vyas population manifolds, Turpin/Lanham faithfulness, graph-constrained decoding, Moschella relative representations — whose full references must be added and verified before submission. They are named, not yet keyed.